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THE DISTRIBUTED LAG BETWEEN CAPITAL APPROPRIATIONS AND EXPENDITURES[1]

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THIS PAPER PRESENTS a new distributed lag, very flexible and easy to estimate, and its application to the problem of predicting quarterly capital expenditures in manufacturing industries from present and past appropriations.[2] One would expect, encouraged by a few simple correlation coefficients, that computing the weighted sum of past appropriations which comprises current expenditures would require a distribution of weights which rise for a time, then decline. Here it is assumed only that successive weights lie on a polynomial. The plan is to estimate as regression coefficients a few points on the curve, taking account of the fact that polynomial interpolation will be used to interpolate between them for the remaining points. The same equation may include other variables with different distributed lags, or unlagged variables. Since the length of the distributed lag is generally not known in advance, it is necessary to estimate the distribution using varying numbers of periods, then choose the best among them.

The quarterly data on appropriations and expenditures come from the survey conducted by the National Industrial Conference Board (NICB) among the thousand largest manufacturing companies. Estimates in this paper are based on the nine years, 1953 through 1961. The distributed lag was estimated for all manufacturing, for durables and nondurables separately, and for their fifteen constituent industries. The distributed lag on appropriations, plus estimated seasonal coefficients, gives a good explanation of the variance in expenditures, with the best lags centering around 8 and 9 quarters in length. The distributions are quite stable, that is, among the several distributions estimated for each industry, the weights for a given period vary little after a sufficiently long distribution has been reached.

The distributed lag will be described first, then the results of its application to the appropriations-expenditures data will be reported. Finally, because one is usually interested in total capital expenditures for the manufacturing sector, rather than the 60 per cent or so accounted for by the large companies reporting to the NICB, we shall look briefly at the relationship between the expenditures reported in the

[1] This paper was read at the September, 1963, Econometric Society meetings. I am grateful to James S. Duesenberry, under whose direction the work was done as a doctoral dissertation, for helpful comments. The computing was done at the MIT Computation Center.

[2] Attempts have also been made to estimate (non-statistically) the rate of spending from applications for accelerated amortization and from another questionnaire designed specifically for this purpose [7, 8]. Kareken and Solow [1] have made statistical estimates of the distributed lag between new orders for nonelectrical machinery and the Federal Reserve Board index of production of new business equipment.

NICB survey and those in the Office of Business Economics-Securities and Exchange Commission (OBE-SEC) surveys.

1. THE DISTRIBUTED LAG

All distributed lag equations state that a dependent variable, Y, is determined by a weighted sum of past values of an independent variable, X:

$$(1) \qquad Y_t = \sum_{i=0}^{n-1} w(i)\, X_{t-i}.$$

If n, the number of relevant values of X is small, as may well be the case for some problems if annual data are involved, and if these successive past observations are not collinear, then the $w(i)$, the weights with which the several present and past values are combined, can be estimated directly by least squares.[3] When n is large, however, or when successive observations are too collinear for this straightforward treatment, as will frequently be the case with quarterly data, it becomes necessary to make some reasonable, restrictive assumption about the pattern of the weights. The point is, of course, to choose an assumption which makes the individual lag coefficients depend on a few parameters, which in turn can be estimated in some reasonably simple way.[4]

The "interpolation distribution" assumes that the $w(i)$ are values at $x=0, \ldots, n-1$ of a polynomial $w(x)$ of degree $q+1$, $q<n$, where n is the number of periods over which the distributed lag extends. Its estimation is based on the fact that once $q+2$ points on the curve are known—$w(x_0)=b_0$, $w(x_1)=b_1, \ldots, w(x_{q+1})=b_{q+1}$—all the $w(i)$ can be calculated as linear combinations of these known values by

$$(2) \qquad w(i) = \sum_{j=0}^{q+1} \Phi_j(i)\, b_j \qquad (i=0, \ldots, n-1).$$

Here the $\Phi_j(i)$ are the values at $x=i$ of the Lagrangian interpolation polynomials

[3] For an example of a distributed lag estimated in this way see **[4]**.

[4] Two assumptions for which estimating procedures have been worked out are that the weights decline geometrically and, a more general form of the same thing, that they are points on the Pascal distribution **[6, 11]**. The first-mentioned seemed not so well suited for the present problem since it was expected that the true distributions might be more or less symmetric. Numerous attempts were made to estimate the Pascal distribution with appropriations-expenditures data for the chemical industry, but estimates of one of the parameters, λ, in the notation of **[11]**, did not fall within the required range, 0 to 1, possibly because the estimation methods are rather sensitive to specification error. The distribution described below developed more directly from a third distributed lag, which assumes that the pattern of the weights is an inverted V **[3]**.

$$\Phi_0(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_{q+1})}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_{q+1})},$$

$$\Phi_1(x) = \frac{(x-x_0)(x-x_2)\dots(x-x_{q+1})}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_{q+1})},$$

$$\vdots$$

$$\Phi_{q+1}(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_q)}{(x_{q+1}-x_0)(x_{q+1}-x_1)\dots(x_{q+1}-x_q)}.$$

Noting that these polynomials have the property that

$$\Phi_j(x_j)=1 \qquad (j=0,\dots,q+1)\,,$$
$$\Phi_j(x_k)=0 \qquad (j\neq k;\, j=0,\dots,q+1;\, k=0,\dots,q+1)\,,$$

it is easily seen that

$$w(x) = \sum_{j=0}^{q+1} \Phi_j(x) b_j$$

is indeed a polynomial of degree $q+1$ having the values b_j at the points x_j as required. Hence equation (2) is justified. Since we shall always want $w(-1)=w(n)=0$, i.e., zero weights before time 0 and after time $n-1$, we may take $x_0=-1$, $x_{q+1}=n$, and $b_0=b_{q+1}=0$. Then equation (2) simplifies to

$$\text{(2a)} \qquad w(i) = \sum_{j=1}^{q} \Phi_j(i) b_j\,.$$

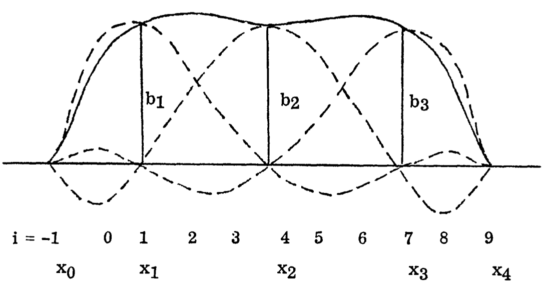

FIGURE 1.—Example of Lagrangian Interpolation.

Figure 1 shows with dashed lines an example of the $\Phi_j(x)$ for $q=3$, $n=9$, and $x_0=-1$, $x_1=1$, $x_2=4$, $x_3=7$, $x_4=9$, assuming that $b_1=b_2=b_3$. The interpolated polynomial $w(x)$, computed from (2a), is drawn with a solid line.

Substituting (2a) into (1) gives

$$\text{(3)} \qquad Y_t = \sum_{i=0}^{n-1} \left(\sum_{j=1}^{q} \Phi_j(i) b_j \right) X_{t-i} = \sum_{j=1}^{q} b_j \sum_{i=0}^{n-1} \Phi_j(i) X_{t-i}\,.$$

The b_j can now be estimated by simply regressing the Y_t on the q variables $z_{tj} = \Sigma_{i=0}^{n-1} \Phi_j(i)X_{t-i}$, $j=1, \ldots, q$. The distributed lag weights are then calculated from (2a). Another distributed lag or unlagged variables could of course be included in (3). The computational steps are as follows:

(*a*) Pick n and q. One will frequently have some idea about how long a lag is relevant for a particular problem, but if further indications are desired, the method used with the appropriations-expenditures problem can be recommended. It was simply to calculate for one industry the simple correlation coefficients between the dependent variable and successive lagged values of the independent variable. That quarter in which appropriations explained expenditures about as well as appropriations with no lag at all was chosen as the center of a range of n's. The plan is to estimate a number of lags of different length, then choose the best among them.

The number of parameter points, q, might be determined to some extent by the number of observations available. In addition to the points estimated, it must be remembered that the two points $(t+1)$ and $(t-n)$ are specified in advance to be zero, so that the polynomial being fitted will always pass through $(q+2)$ points.

(b) Choose the location of the q points x_j. It makes no difference where in the interval $[0, n]$ these parameter points are located. The intuitive explanation is that within this interval there can be only one polynomial of degree $(q+1)$ which minimizes the sum of squares. The parameter points do not need to be integers, and since it is most convenient to choose them in standard form, they usually will not be. As a rule, therefore, all the distributed lag weights will be calculated as linear combinations of the q estimated ones.

(c) Calculate the Lagrangian interpolation coefficients $\Phi_j(i)$, as indicated above.

(d) Compute $z_{tj} = \Sigma_{i=0}^{n-1} \Phi_j(i)X_{t-i}$ for all $t \geqslant n$ and $j=1, \ldots, q$. This transformation of the independent variable expands it into q variables. In time series, it also reduces the number of observations by n—1.

(e) Use multiple regression to estimate the b_j for

$$(4) \qquad Y_t = \sum_{j=1}^{q} b_j z_{t_j} + u_t .$$

(f) Use equation (2a) to compute $w(i)$, $i=0, 1, \ldots, n-1$. The standard errors of the distributed lag weights, combining both variances and covariances of the estimated regression coefficients, are calculated by

$$(5) \qquad s_i^2 = \phi_i \sigma^2 (Z'Z)^{-1} \phi_i' ,$$

where $\phi_i = (\Phi_1(i), \ldots, \Phi_q(i))$ and $\sigma^2(Z'Z)^{-1}$ is the variance-covariance matrix of b_j.

(g) Vary n over the range already chosen and repeat steps (a) through (f).

Ways to choose the best lag distribution from among the several estimated will be discussed in connection with the use of the model below.

2. THE DATA

The NICB's quarterly survey among the thousand largest manufacturing companies covers appropriations made during the quarter, the appropriations backlog at beginning and end of the quarter, appropriations committed (i.e., orders placed) during the quarter, cancellations of previously made appropriations, and capital expenditures. The information requested on the last mentioned item is identical with that collected in the OBE-SEC quarterly survey. Appropriations are thought of as the final stage of approval for capital expenditures, a confirmation of plans represented in the annual budget.[5] They are data routinely tabulated by companies using formal capital budgeting procedures. The NICB publishes (in the *Conference Board Business Record*) results of the survey by 2-digit SIC industries. Since 1956, for all manufacturing and for total durables and nondurables, it has also blown up the replies of the reporting companies to the level of the thousand largest companies, using total assets in three different size groups as the basis of inflation.

When the survey was begun in 1956 about 500 companies replied with data for 1955 and 1956. These respondents represented 69 per cent of total assets in the survey's universe, and 55 per cent of employment. The response rate by industry varied from 95 per cent of assets, in primary iron and steel, to 40 per cent, in transportation equipment. Of the first 500 respondents, 353 were able to provide data for 1953 and 1954, and these were linked by the NICB to the 500-company series, which ran from 1955 through 1959.

Since the beginning of 1958 respondents have included 602 companies, owning 80 per cent of total assets among the thousand largest companies. I used the 8 quarters overlap in 1958–59 to link the earlier series to the 602-company level. For appropriations (A), for example, each quarter's appropriations prior to 1958 were multiplied by the ratio

$$\frac{\sum_{t=1}^{8} A_t \text{ (602 companies)}}{\sum_{t=1}^{8} A_t \text{ (500 companies)}}.$$

During 1958, the differences between the 500-company series thus inflated to the 602-company level and the actual 602-company series were divided, with gradually increasing weight to the latter. Since the first quarter of 1959 (1959–I) the 602-company series has been used as reported. Expenditures and cancellations were linked in the same way.

[5] For further discussion of the relevant aspects of capital budgeting and the NICB survey see **[9]** and **[10]**.

3. THE MODEL

The basic assumption underlying the following analysis of the appropriations-expenditures data is that expenditures accrue entirely from previous appropriations, that no capital expenditure is made without an appropriation, and that all appropriations are eventually spent. In other words, the economic decision to invest is made at the appropriation stage, and thereafter spending is determined by technological and institutional factors.[6] Spending of an appropriation is assumed to begin in the same quarter in which it is made, and the problem now is to determine what part of it is spent in this and each succeeding quarter. Cancellations must be mentioned, since companies reported that they cancelled appropriations, in the period 1953 through 1961, amounting to 6.7 per cent of total appropriations made. Not to adjust in some way for cancellations will cause the distributed lags to subtract the average cancellation in every quarter, whereas in fact cancellations have been quite erratic. An attempt to take account of cancellations will be described, but the conclusion was that it is usually not worth the effort.

It was early decided to use a three-parameter distribution, i.e., to estimate fourth-degree polynomials to describe the pattern of spending of appropriations. In some sense, n, the number of periods in the distribution, must be considered a fourth parameter, since its optimal value is routinely determined by trying alternative values. The range of n's to be tried was arrived at, as noted above, by computing the simple correlation coefficients between expenditures and successive lagged values of appropriations in the chemical industry, both for levels of these two series and for first differences. The correlation coefficients for levels ran .08, .45, .68, .82, .86, .77, .56, .36, and .09. First differences yielded positive coefficients of more erratic sizes until the ninth quarter back, which was slightly negative. On the basis of these numbers it was expected that the optimal length of lag for this industry was about nine quarters. Accordingly I decided to try $n=6, 7, \ldots, 12$ for all industries.

Both appropriations and expenditures for manufacturing as a whole have a seasonal pattern, and, although some industries' appropriations do not, it was clearly necessary to make some seasonal adjustments. Rather than correcting the series before estimating the distributed lag between them, three seasonal dummy variables were included in the estimating equation, the fourth substituted for by defining it to be the number necessary to make the sum of the four equal to zero, in accordance with the assumption of a zero intercept for the year as a whole. The six-variable equation to be estimated by multiple regression then is

$$E_t = a_1 s_1 + a_2 s_2 + a_3 s_3 + \sum_{j=1}^{3} b_j \sum_{i=0}^{n-1} \Phi_j(i) A_{t-i} + u_t \tag{6}$$

[6] This approach shifts the more interesting problem, so far as economic theory is concerned, to the study of appropriations. Work using appropriations as the dependent variable is under way and will be reported later.

for $n=6, 7, \ldots, 12$, where E is expenditures; s_1, s_2, s_3 are seasonal dummies; A, appropriations; and the remainder of the terminology is as defined in the previous discussion of the distributed lag. The parameter points x were located at $-.6$, 0, $+.6$, in the standardized interval, with -1 and 1 defined in advance to be zero.[7] Equation (6) then was estimated for all manufacturing, for durable and nondurable goods industries separately, and for their fifteen constituent industries.

4. EMPIRICAL RESULTS

From among the 7 distributed lags estimated for each industry, one has been chosen as the best, using two rules of thumb and a modicum of judgment. These distributions explain appropriations rather well. The coefficients of multiple determination adjusted for number of observations ($\bar{R}^2$) in 12 of the 15 industries are greater than .8, and for the aggregates, all manufacturing, durables, and nondurables, they are above .9.

We first present some observations on the sets of distributed lags estimated for each industry, although, in general, the empirical results will be presented first for all manufacturing and then for the constituent industries. The first but not the final criterion to be used in choosing the best distributed lag is $\bar{R}^2$. In this respect there frequently is not much difference between the worst and the best distributions. In the 18 industries and aggregates thereof, the gap runs from .02 to .32. For all manufacturing, it is .04; for most industries, less than .1. In general, the correlation coefficients do not bounce about from one distributed lag to the next, but move slowly up to a peak, and then decline.

With these usually slight differences in degree of explanation of the dependent variable offered by successive distributed lags, it is not surprising that consecutive distributions do not change much either, especially after a sufficiently long one has been reached. In most industries, the shorter distributions are the least satisfactory, giving the lowest correlation coefficients, and frequently having kinks and jagged peaks. Usually by the time 8 or 9 quarters are included, both these characteristics disappear and a smooth curve takes shape. Thereafter, most of the weights of succeeding lag distributions fall within one standard error of the best one for the corresponding period. A concomitant of this behavior in the early, "relevant" part of the longer-than-optimal distributions is that the later weights frequently become negative, since the polynomial cannot, of course, continue at zero. All manufacturing is a typical, if subdued, example of the changes that occur as the

[7] I didn't realize until I had finished computing that the location of the parameter points was a matter of indifference. The ones I used were chosen according to the theorem in numerical analysis which says that the way to minimize the maximum possible deviation of an interpolated polynomial from the true function, providing the latter has n continuous derivatives and the nth derivative is differentiable, is to pick as the parameter points the roots of the Chebyshev polynomial.

number of periods in the distributed lag moves from 6 to 12, and its 7 lag distributions are shown in Figure 2. The highest $\bar{R}^2$ is reached with the 8-period distribution, and this is also the one where relative stability in the weights appears. Of the first 8 weights of distributions with 9 through 12 periods, all (except one in the longest distribution) fall within one standard error of the corresponding weights in the optimal 8-period distribution. In the 9-period distribution, all the weights are still positive, but beginning with 10 periods, the last one and then two are negative by 1 or 2 per cent. In view of the slight differences sometimes found in the correlation coefficients, close similarity of the weights among distributed lags of optimal and longer length may be considered a second criterion for choosing the optimal distribution.

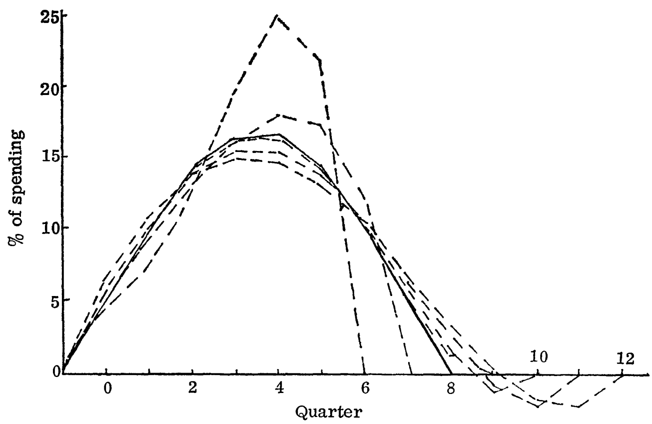

FIGURE 2.—Distributed Lags on Appropriations, All Manufacturing.

It should be noted here that the sum of the distributed lag weights is substantially the same from the shortest to the longest distributions. The individual weights of the shorter distributions are simply larger, as seen in Figure 2. For all manufacturing, for example, the weights always add to .92. An additional 7 per cent of appropriations is accounted for by cancellations, and the remaining 1 per cent may possibly be due to lack of reporting on cancellations.

The weights of the optimal 8-period distribution for all manufacturing are also shown in the first column of Table I. They are roughly symmetric, 15 per cent of appropriations being spent within the first half year, 45 per cent in a year, 77 per cent during the first 6 quarters and, as noted above, 92 per cent by the end of 2 years. This distribution yields a coefficient of multiple determination of .920. Expenditures predicted using it and the estimated seasonal coefficients are shown in Figure 3 against actual expenditures. Appropriations are also shown. All turning points are correctly predicted. The residuals range from virtually zero to a little

TABLE I

ALL MANUFACTURING, BEST DISTRIBUTED LAGS

		1	2	3
		Explanatory variables including		
		Appropriations Only	Appropriations with Backlog	Appropriations Backlog. Cancellations
Seasonal intercepts, for quarters (million $)	I	−283.	−311.	−318.
	II	13.	6.	34.
	III	− 50.	− 42.	− 47.
	IV	320.	347.	330.
Distributed lag weights, for quarters	0	.048	.068	.066
		(.023)	(.023)	(.018)
	1	.099	.122	.113
		(.016)	(.017)	(.016)
	2	.141	.156	.141
		(.013)	(.013)	(.010)
	3	.165	.168	.153
		(.023)	(.021)	(.014)
	4	.167	.157	.151
		(.023)	(.022)	(.018)
	5	.146	.127	.136
		(.013)	(.014)	(.016)
	6	.105	.084	.111
		(.016)	(.017)	(.011)
	7	.053	.037	.078
		(.024)	(.022)	(.014)
	8			.040
				(.016)
Regression Coefficient for backlog dummy			−113.	− 76.
			(48.)	(47.)
Sum of weights		.922	.919	.990
Cancellations/Appropriations		.067	.067	.067
$\bar{R}^2$		.920	.933	.940
Durbin-Watson Statistic		.890	1.409	1.314

over 11 per cent of actual expenditures, averaging about 4 per cent. Small as they are on the whole, it is of some interest to examine them further.

From 1954–IV through 1955–II expenditures are a little larger than predicted. The investment boom of 1956–57 is characterized first by a period when actual expenditures fell short of predicted, from 1955–IV through 1957–II, followed by a burst of overspending, in the last 2 quarters of 1957. (But the underspending in the first 7 quarters of this period totaled more than twice the sum which was eventually "made up" in the last half of 1957.) Overspending was cut short at the beginning of

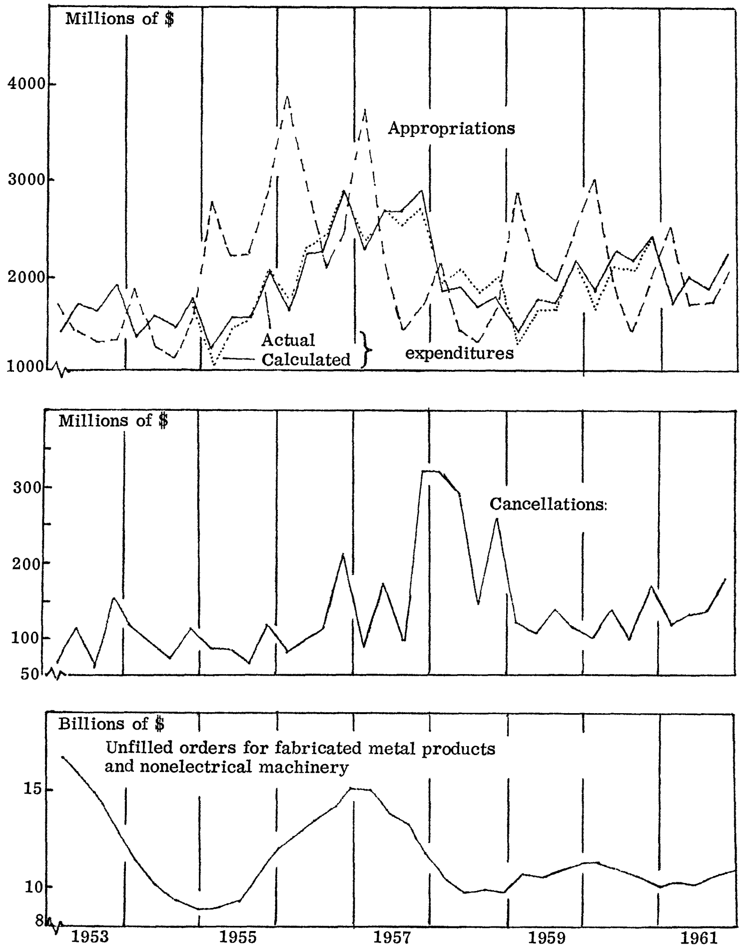

FIGURE 3.—Appropriations, Expenditures, and Related Series for All Manufacturing.

1958, when in every quarter spending fell short of predicted. From 1959 through 1961 spending was larger than predicted from appropriations, by amounts ranging from virtually zero to nearly 7 per cent of spending.

This summary suggests that the residuals from equation (6) are autocorrelated. That they are is confirmed by the Durbin–Watson statistic, which is .9, indicating definite autocorrelation. Autocorrelation here is not terribly serious, since it does not mean biased estimates, and the residuals are quite small. It only indicates that the standard errors of regression coefficients (and indirectly of the distributed lag weights) are understated. The regression coefficients, however, are substantially more than twice their standard errors (for all manufacturing the ratios are 4.9, 6.7, and 5.1), and at any rate no decisions about the usefulness of variables have been made on those grounds. Nevertheless, since two of the most important runs of

TABLE II

BEST DISTRIBUTED LAGS FOR INDUSTRIES

		1 All Manu-facturing	2 All Durable Goods	3 Iron and Steel	4 Non-ferrous Metals	5 Electri-cal Ma-chinery	6 Machinery Except Electrical	7 Trans-portation Equipment	8 Stone, Clay and Glass
Seasonal intercepts,	I	−311.	−158.	−45.4	−10.7	−36.9	−10.9	−32.5	−15.3
for quarters	II	6.	− 2.	8.8	− 3.3	− 5.9	− 3.0	2.2	2.6
(million $)	III	− 42.	− 28.	−22.9	− 2.5	3.2	− 2.5	3.9	− 1.0
	IV	347.	188.	59.6	16.5	39.5	16.4	26.5	13.7
Distributed lag	0	.068	.057	.013	.076	.072	.064	.058	.061
weights, for quarters		(.023)	(.021)	(.030)	(.023)	(.019)	(.023)	(.016)	(.021)
	1	.122	.101	.047	.112	.110	.127	.119	.114
		(.017)	(.016)	(.024)	(.023)	(.018)	(.018)	(.014)	(.017)
	2	.156	.132	.092	.123	.124	.169	.166	.150
		(.013)	(.014)	(.021)	(.014)	(.010)	(.014)	(.016)	(.010)
	3	.168	.151	.138	.119	.123	.181	.186	.166
		(.021)	(.021)	(.029)	(.009)	(.012)	(.020)	(.020)	(.017)
	4	.157	.156	.175	.109	.113	.161	.177	.160
		(.022)	(.020)	(.028)	(.016)	(.019)	(.021)	(.016)	(.022)
	5	.127	.145	.190	.097	.099	.114	.140	.136
		(.014)	(.013)	(.019)	(.020)	(.020)	(.015)	(.011)	(.017)
	6	.084	.118	.174	.086	.084	.056	.086	.097
		(.017)	(.019)	(.026)	(.018)	(.014)	(.018)	(.020)	(.010)
	7	.038	.071	.114	.076	.068	.007	.032	.053
		(.023)	(.024)	(.033)	(.013)	(.010)	(.021)	(.024)	(.017)
	8				.063	.052			.016
					(.013)	(.015)			(.020)
	9				.040	.031			
					(.013)	(.017)			
Regression coefficients	BLC	−113.	− 55.	−30.3	−24.5	14.6	− 2.6	−12.7	
for backlog dummy		(48.)	(35.)	(16.0)	(8.5)	(5.4)	(3.1)	(13.9)	
Sum of weights	SW	.919	.931	.944	.903	.875	.879	.964	.952
Cancellations/Approp.	C/A	.067	.057	.017	.073	.121	.089	.067	.052
$\bar{R}^2$	$\bar{R}^2$	.933	.922	.844	.915	.863	.839	.906	.886
Durbin-Watson statistic	DW	1.396	1.487	1.631	1.015	.900	1.557	.903	1.631

[1] Includes lumber products, furniture and fixtures, and miscellaneous manufactures.

autocorrelated residuals correspond to the behavior of variables whose influence had been suspected earlier, these variables were now added, one at a time, to equation (6).

The obvious explanation for expenditures falling short of predicted during 1956–57, and the compensatory overspending in the last half of 1957, is the tem-

	9 Fabricated Metal Products	10 Other[1] Durables	11 All Non-durable Goods	12 Food and Beverages	13 Tex-tiles	14 Paper	15 Chemi-cals	16 Petro-leum	17 Rubber	18 Other Non-dur.[2]
I	−7.38	−5.08	−176.	−11.4	−3.62	12.7	−39.9	−68.	−6.59	−1.44
II	− .08	−2.00	6.	6.7	.67	−13.5	.5	−22.	.41	.75
III	−1.77	−1.21	− 5.	1.1	1.77	− .3	− 2.0	−27.	−1.14	−1.36
IV	9.24	8.29	175.	3.6	2.18	1.1	41.4	117.	7.32	2.05
0	.085	.081	.126	.011	.022	.102	.063	.184	.103	.120
	(.042)	(.042)	(.021)	(.052)	(.033)	(.022)	(.022)	(.044)	(.018)	(.060)
1	.153	.137	.174	.105	.155	.148	.103	.270	.136	.130
	(.038)	(.028)	(.017)	(.041)	(.021)	(.021)	(.018)	(.027)	(.016)	(.043)
2	.191	.169	.172	.200	.266	.159	.127	.247	.130	.115
	(.035)	(.026)	(.009)	(.037)	(.031)	(.014)	(.011)	(.032)	(.011)	(.045)
3	.195	.176	.144	.245	.285	.148	.138	.144	.111	.121
	(.041)	(.037)	(.016)	(.045)	(.029)	(.017)	(.019)	(.030)	(.015)	(.057)
4	.166	.161	.108	.221	.201	.128	.138	.027	.093	.157
	(.037)	(.031)	(.021)	(.039)	(.022)	(.020)	(.024)	(.040)	(.018)	(.044)
5	.113	.124	.075	.139	.068	.105	.130		.084	.197
	(.030)	(.027)	(.018)	(.043)	(.039)	(.018)	(.019)		(.015)	(.047)
6	.051	.070	.051	.041		.083	.113		.084	.177
	(.046)	(.036)	(.011)	(.054)		(.015)	(.012)		(.012)	(.064)
7	.004	.	.035			.062	.087		.083	
	(.052)		(.014)			(.020)	(.019)		(.017)	
8			.022			.037	.051		.064	
			(.018)			(.021)	(.023)		(.018)	
9										
BLC	−5.45	3.03	− 49.	− 1.68	−3.82	−10.0		− 7.2		
	(3.25)	(1.72)	(17.)	(2.69)	(1.11)	(4.3)		(8.2)		
SW	.958	.918	.908	.962	.998	.972	.950	.872	.888	1.017
C/A	.037	.048	.074	.043	.047	.046	.064	.091	.056	.010
$\bar{R}^2$	.647	.855	.945	.782	.853	.870	.850	.883	.819	.596
DW	1.640	1.582	1.556	.978	1.554	1.300	.701	1.636	1.096	.718

[2] Includes tobacco, apparel, printing and publishing, and leather products.

porary limitation in supply of capital goods, caused by heavy demand and aggravated by the steel strike in 1956. It is reflected in the end-of-quarter backlog of unfilled orders for fabricated metal products and nonelectrical machinery, which is shown at the bottom of Figure 3.

It was decided to take account of this supply constraint by including a variable which would be $+1$ during the 5 quarters beginning in 1955–IV, -1 in the following 5 quarters, and zero elsewhere. Equation (6) then becomes

$$E_t = a_1 s_1 + a_2 s_2 + a_3 s_3 + a_4 S + \sum_{j=1}^{3} b_j \sum_{i=0}^{n-1} \Phi_j(i) A_{t-i} + u_t , \tag{7}$$

where S is the dummy variable representing the supply constraint. Expenditures were then regressed on these 7 variables for $n = 6, \ldots, 12$ in the same 18 industries and aggregates as before.

The effect of the additional variable on the best distributed lag for all manufacturing can be seen in Column 2 of Table I. The estimated regression coefficient appears with the expected negative sign, it makes a significant contribution by the usual standards, and, as hoped, the autocorrelation in the residuals has decreased. The Durbin-Watson statistic has risen from .9 to 1.4. The best distributed lag is again 8 quarters long, the weights still add to about .92, and individually are little changed, the maximum difference being about 2 per cent. All the new weights fall within 2 standard errors of the original ones, and 4 of the 8 are within one standard error. Most important, the general effect of the backlog variable is to shift the distribution forward, i.e., it raises the first 4 weights and lowers the last 4, indicating that the new variable does indeed compensate for the delayed spending. Incidentally, $\bar{R}^2$ rises slightly, from .920 to .933. As for the effect on the residuals 1955–IV through 1958–I, the overestimate of spending is reduced in each of the first 5 quarters, though not until 1956–IV is the small overestimate switched to a larger underestimate. In the last half of the 10-quarter period involved, when the backlog of orders was being worked down, the original predictions were underestimates only in the last 2 quarters of 1957. Taking account of the backlog of orders does reduce these 2 errors to about half their former size, but does not change the sign of any of the 5. In short, the new variable produces changes in the expected direction, but does not completely erase the spell of autocorrelated residuals.

Besides 1956–57, the other period when there is an obvious explanation for deviation of predicted from actual spending is in 1958, when expenditures were a good deal less than predicted. Here the underspending of a predicted lower level of investment appears to be voluntary. Cancellations shot up in 1957–II and remained high in 3 of the 4 quarters in 1958. They too are shown in Chart 2. Comparison of these cancellations with the residuals from equation (7) reveals no general comovement, except in 1958. Nevertheless, since some adjustment for cancellations had been seriously considered before any computing was done, but delayed over the matter of which appropriations are cancelled, an attempt to answer that question was now made. Expenditures predicted by equation (7) were multiplied by the reciprocal of the sum of weights shown in Column 2 of Table 1, and the residuals were recalculated. The new residuals, almost all of them positive, were now used as a dependent variable to be explained by cancellations, lagged successively from zero to three quarters. The four simple correlation coefficients thus obtained were, respectively, .264, .122, .197, and $-.022$. None of these is significantly different from zero at the .95 confidence level, and the consideration of cancellations might have been dropped at this point.

Having come this far, however, the decision was made to re-estimate equation (7) after adjusting the data for cancellations. Cancellations in each quarter were divided by three, and one-third added to expenditures in the current and each of the following 2 quarters. Equation (7) was re-estimated with this data, but as a

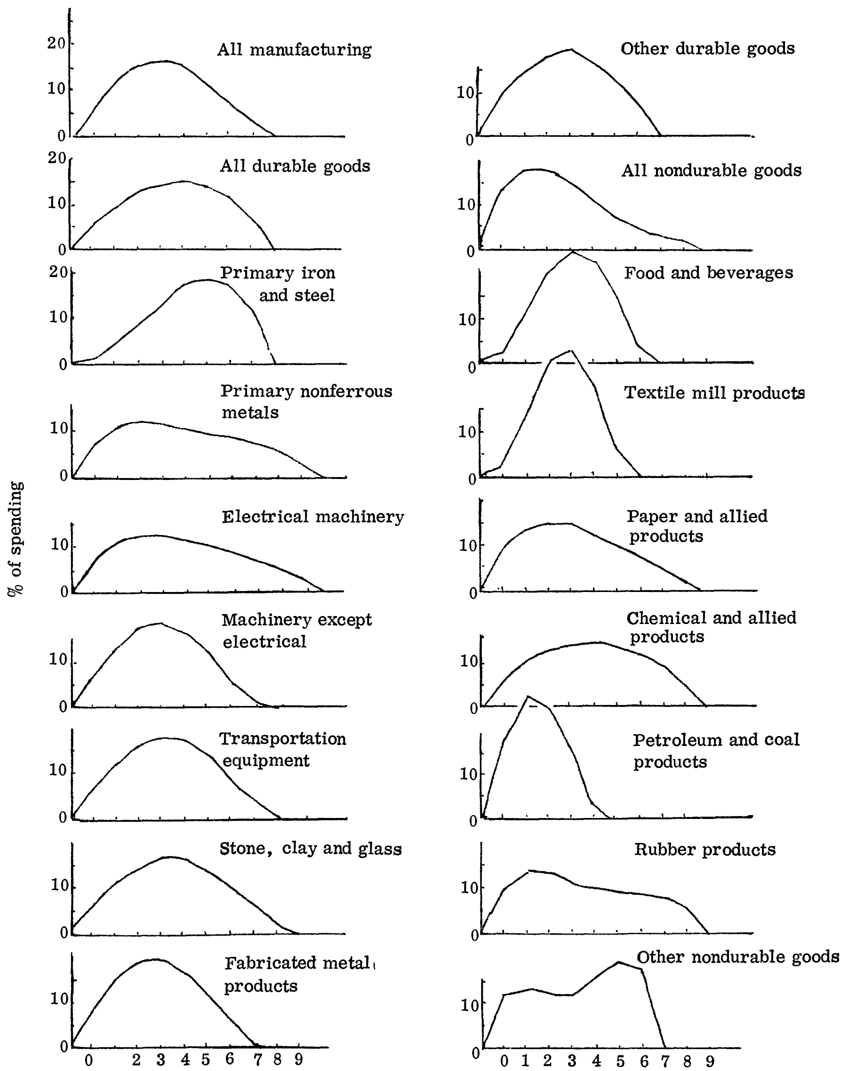

FIGURE 4.—Best Distributed Lags.

final step, cancellations were subtracted in exactly the same way they had earlier been added in. The results of this adjustment are shown in Column 3 of Table I. As expected, the distributed lag weights now add to .99. The coefficient of multiple determination attached to the 9-quarter distribution is greater than that for the 8-quarter one (by .005), and is slightly greater than either of the two previous best $\bar{R}^2$. The regression coefficient for the backlog-of-orders variable is no longer twice its standard error, and autocorrelation of the residuals, as measured by the Durbin-Watson statistic, is slightly greater than before the cancellations adjustment was made. All in all, the adjustment for cancellations, unlike that for the supply constraint in 1956–57, is felt to be of limited usefulness, at least in periods like 1961 and the first half of 1962 when cancellations were about average. If predicting in a period of unusually high (or low) cancellations, like 1958, however, some adjustment might be worthwhile.

Equations (6) and (7) were estimated for individual industries as for all manufacturing. The inclusion of the backlog variable produced less autocorrelation in 11 of the 15 industries, and slight increases in the remaining 4. This lack of improvement in 4 cases is not surprising in view of the fact that residuals resulting from equation (6) show that all industries were not affected as was manufacturing as a whole in 1956–57. For the 4 industries in which the additional variable yielded no reduction in autocorrelation, the distributed lags presented in Figure 4 and Table II are those produced by equation (6). The other distributed lags come from equation (7). The cancellations adjustment caused slightly more autocorrelation in 7 industries, slightly less in 8, and none of the results presented below are based on those calculations.

In 4 of the 15 industries the best distributed lag, chosen by the correlation coefficient criterion, contained negative weights, as did the longer distributions for all manufacturing. The obvious and simple solution to this problem[8] in 3 of the 4 industries was to move back 1 or 2 periods to a distribution whose weights were all positive, and in each case the weights became stable with these shorter distributions. In the petroleum and coal products industry, every distributed lag, 6 through 12 periods long, had negative weights. The indication here is that even 6 periods is too long, though the highest $\bar{R}^2$ was attached to the 12-period distribution. Distributions 3, 4, and 5 periods long were estimated for this industry, and with 5

[8] It is possible to "urge" the distributions to remain zero rather than becoming negative beyond their endpoints by adding another parameter point, specified in advance to equal zero, at a distance ε beyond the endpoint at which negative weights occur. If then $\varepsilon \to 0$, the result is simply to repeat the last (or first) term of the formula for the Lagrange weights. If one additional point does not suffice to produce all positive weights in the distribution with highest $\bar{R}^2$, more can be added. The effect of one additional point is to specify a zero first derivative at the endpoint; another point produces a zero second derivative as well, etc. Trying this approach with 2 industries produced good results in the case where the negative weights were small to begin with, i.e., not significantly different from zero; but even 4 extra points did not succeed in eradicating negative weights in the distribution with highest $\bar{R}^2$ when they were more than twice their standard errors to begin with.

periods $\bar{R}^2$ reaches a local maximum, all the weights are positive, and furthermore the first 5 weights of each succeeding distribution are very close to the weights of the 5-period one. It is the one included here.[9]

Table II and Figure 4 present then the best distributed lags between appropriations and expenditures, for manufacturing as a whole, durables and nondurables separately, and for 15 individual manufacturing industries. In 8 of the industries, appropriations are spent over 8 or 9 quarters, 5 spend them in less than 2 years, and 2 industries, in $2\frac{1}{2}$ years. The single period in which the greatest expenditures are made, i.e., the mode of the distributions, varies from the second to the sixth. For about half the industries, as for manufacturing as a whole, it is the fourth period. The half-way point in the spending of all appropriations made is found in 6 industries, as for manufacturing as a whole, between the fourth and fifth quarters; in 7 more industries, between the third and fourth. At the extremes, iron and steel companies have not spent half their appropriations until after the fifth quarter, while petroleum companies have spent 45 per cent after 2, and 70 per cent after 3 quarters. These two extreme cases seem to be largely responsible for skewing durable goods investment spending to the left, nondurables to the right.

Though an explanation of why spending patterns differ as they do among industries lies outside the scope of this paper, a rough check was made on one of the first possible reasons to come to mind, the proportion of spending for plant or new construction *vs.* equipment. Since the beginning of 1959, 464 companies in 14 industries (the breakdown is not feasible in petroleum) have been supplying to the NICB information on how much of their appropriations are for plant, how much for equipment. The coefficient of rank correlation between the average share of appropriations for plant over the 3 years and, as a measure of the rate of spending, the percentage of appropriations which have been spent after one year was computed, and found to be .12. The main conclusion to be drawn from this result, however, should probably be that uncovering the causes of variation among industries will require a more subtle approach.

Results of Aggregation

It has been noted before that the degree of explanation of the variance in expenditures is higher for the aggregates, all manufacturing, durables and nondurables, than for the individual industries. $\bar{R}^2$ for all manufacturing does lie between the $\bar{R}^2$ for its two components, but all three of these aggregate determination coefficients are higher than any of those for individual industries. This result is not surprising in view of the rule explained by Grunfeld and Griliches [5], that the higher the correlation between the independent variables of different behavior

[9] The short distribution in petroleum is consistent with the fact that this industry, more than any other, appropriates on an annual basis. Forty-six per cent of petroleum appropriations over the 9 years were made in the first quarter. In all remaining industries, the comparable figure is only 27 per cent.

units, *ceteris paribus*, the higher the R^2 of the aggregate equation relative to the R^2 of the micro equations.

In the context of this study it is easy to imagine that the most disaggregated series, i.e., the industries, might be of interest per se. Nevertheless, if one were interested only in predicting for all manufacturing, one might well ask what is to be gained by estimating 2 equations, durables and nondurables, or the 15 industry equations, rather than simply one, for the aggregate. To answer this question, we added residuals algebraically over durables and nondurables (2 regressions) and over the 15 industries, noting subtotals for durables and nondurables; we then computed the sums of squares (Σu^2) of these 4 series of residuals, along with the same figure for the residuals of 3 individual regression equations: all manufacturing, durables, and nondurables.

TABLE III

RESULTS OF AGGREGATION

Level of Aggregation		Σu^2	R^2
All manufacturing (one equation)		20,401,200	.94703
Durables and nondurables (two equations)		20,001,933	.94807
Individual industries (fifteen equations)		17,984,117	.95331
Durables (one equation)	8,304,407		.93854
Durables by industries (eight equations)	7,577,665		.94392
Nondurables (one equation)	3,128,358		.95752
Nondurables by industries (seven equations)	3,193,526		.95663

Table III shows that there is in fact a reduction in Σu^2 of about 2 per cent as one disaggregates from 1 to 2 equations and of about 10 per cent more from 2 to 15. The effect on $\bar{R}^2$ is easily computed, by substituting in the equation

$$R^2 = 1 - \frac{\Sigma u^2}{\Sigma(E-\bar{E})^2},$$

where E is expenditures, the Σu^2 resulting from the algebraic sums of residuals. They are shown in Column 2 of Table III. The result is a definite but small increase as one goes from 1 to 2 to 15 estimated equations. We can also look at the effect of disaggregating from 1 to 8 equations in the case of durable goods, and from 1 to 7 for nondurables. In the former, there results a slight improvement in R^2, in the latter, an even smaller deterioration. This one instance where the aggregate yields a better prediction than the sum of its parts reminds us, however, that there is such a thing as an aggregation gain.[10] In summary, one who was interested only in the aggregate would probably conclude that in this case disaggregation is not worth the effort.

[10] The reasons why an aggregation gain might occur are discussed in some detail in [**5**].

Comparison of NICB and Commerce-SEC Data

As soon as one begins to think of using these distributed lags for forecasting capital expenditures, the question arises of how the NICB sample from the thousand largest companies compares with all manufacturing, as represented by the Commerce-SEC survey. Since companies are asked to report the same figure for capital expenditures in both surveys, we can calculate the ratio between the two series, quarter by quarter. Figure 5 shows spending by the 602 NICB respondents

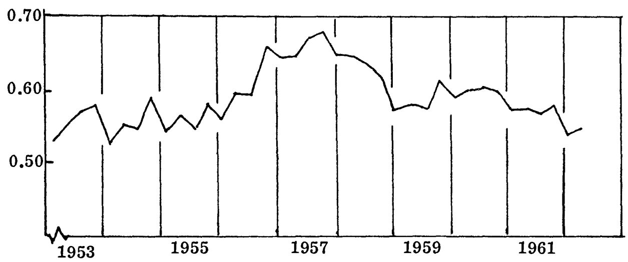

FIGURE 5.—Capital Expenditures: NICB Sample/OBE-SEC Sample.

relative to all manufacturing companies, neither series seasonally adjusted, from 1953 through the second quarter of 1962. The ratio has fluctuated from about .52 to .68, with a seasonal pattern, the cause of which is not obvious. By far the most substantial nonseasonal variation comes during the investment boom centered in 1957, when the large (NICB) companies were accounting for a substantially larger share of total capital spending than they had before or have since. The explanation could be either that it was they who wanted to spend at this time, or that capital goods producers allocated scarce output to their larger customers.

The relative size of industries is naturally different in the two surveys. A comparison [2] for 1957 shows that the proportion of durable and nondurable goods industries is about the same, but within these two large groups differences run as high as 7 per cent. Any careful attempt to use the 602 large companies to predict for all manufacturing would experiment with inflating expenditures by individual 2-digit industries. Working only with the aggregate, since 1959, say, the error resulting from using the ratio between the NICB and the Commerce-SEC series for the same quarter a year ago would average between 1 and 2 per cent.

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